DOM Manipulation in JavaScript - Assignment Solutions

PW EARNERS : Coding and Web Development

By : Nishant Saini Sir

LEC - 39

Section 1 - Selecting and Modifying Elements

1. Select an Element by ID

Question:
Create a heading with an id of title and use getElementById() to select it and change its text.
HTML:
<h1 id="title">Welcome</h1>

Solution:
let heading = document.getElementById("title");
heading.textContent = "Hello JavaScript";

Output:
Hello JavaScript

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2. Select an Element Using querySelector()

Question:
Create a paragraph with a class description and use querySelector() to select it and change its text content.
HTML:
Old Text

Solution:
let paragraph = document.querySelector(".description");
paragraph.textContent = "New Description";

Output:
New Description

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3. Select Multiple Elements Using querySelectorAll()

Question:
Create three <li> elements with the class item. Use querySelectorAll() to select all of them and change their text color using the style property.
HTML:

```
<ul>
   <li class="item">HTML</li>
   <li class="item">CSS</li>
   <li class="item">JavaScript</li>
</ul>
```

Solution:

```
let items = document.querySelectorAll(".item");
items.forEach(function(item) {
    item.style.color = "blue";
});
```

Output:

All three list items will have their text color changed to blue.

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4. Change Content Using textContent

Question:

Create a paragraph containing some text and use textContent to replace its content with a new message.

HTML:

```
<p id="message">Old Message</p>
```

Solution:

```
let message = document.getElementById("message");
message.textContent = "Welcome to JavaScript!";
```

Output:

Welcome to JavaScript!

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5. Add HTML Using innerHTML

Question:

Create a with an id of container and use innerHTML to add a heading and a paragraph inside it.

HTML:

```
<div id="container"></div>
```

Solution:

```
let container = document.getElementById("container");
container.innerHTML = `
    <h2>My Website</h2>
    <p>Welcome to my website!</p>
`;
```

Output:

My Website
Welcome to my website!

Section 2 - Attributes, Classes and Styles

6. Change an Attribute Using setAttribute()

Question:
Create an image element and use setAttribute() to change its src and alt attributes.
HTML:
<img id="myImage" src="old.jpg" alt="Old Image">

Solution:
```
let image = document.getElementById("myImage");
image.setAttribute("src", "new.jpg");
image.setAttribute("alt", "New Image");
```

Output:
The image will now use new.jpg as its source and "New Image" as its alt text.

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7. Add and Remove Classes Using classList

Question:
Create a button and use classList.add() to add a class to it. Then use classList.remove() to remove the class.
HTML:
<button id="myButton">Click Me</button>

Solution:
```
let button = document.getElementById("myButton");
button.classList.add("active");
button.classList.remove("active");
```

Output:
The active class is first added to the button and then removed.

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8. Modify Element Style

Question:
Create a heading and use the style property to change its color, fontSize, and backgroundColor.
HTML:
<h1 id="heading">JavaScript</h1>

Solution:

```
let heading = document.getElementById("heading");
heading.style.color = "blue";
heading.style.fontSize = "40px";
heading.style.backgroundColor = "lightgray";
```

Output:

The heading will display with the new color, font size, and background color.

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9. Read Data Using dataset

Question:

Create a button with a custom data-id attribute and use the dataset property to read its value.

HTML:

```
<button id="productBtn" data-id="101">View Product</button>
```

Solution:

```
let button = document.getElementById("productBtn");
console.log(button.dataset.id);
```

Output:

101

Section 3 - Creating and Adding Elements

10. Create an Element Using createElement()

Question:

Create a new <p> element using createElement(), add some text to it using textContent, and display it on the webpage.

HTML:

```
<div id="container"></div>
```

Solution:

```
let paragraph = document.createElement("p");
paragraph.textContent = "This paragraph was created using JavaScript.";
let container = document.getElementById("container");
container.appendChild(paragraph);
```

Output:

This paragraph was created using JavaScript.

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11. Add an Element Using appendChild()

Question:
Create a <ul> in HTML. Use JavaScript to create a new <li> element and add it to the list using appendChild().
HTML:

```
<ul id="list">
   <li>HTML</li>
   <li>CSS</li>
</ul>
```

Solution:

```
let list = document.getElementById("list");
let newItem = document.createElement("li");
newItem.textContent = "JavaScript";
list.appendChild(newItem);
```

Output:

```
HTML
CSS
JavaScript
```

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12. Add Elements Using append() and prepend()

Question:
Create a list and use append() to add an item at the end and prepend() to add an item at the beginning.
HTML:

```
<ul id="list">
   <li>CSS</li>
   <li>JavaScript</li>
</ul>
```

Solution:

```
let list = document.getElementById("list");
let firstItem = document.createElement("li");
firstItem.textContent = "HTML";
let lastItem = document.createElement("li");
lastItem.textContent = "React";
list.prepend(firstItem);
list.append(lastItem);
```

Output:
HTML
CSS
JavaScript
React

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13. Insert an Element Using insertBefore()

Question:
Create a list containing three items and use insertBefore() to insert a new item before the second item.
HTML:

```
<ul id="list">
    <li>HTML</li>
    <li>JavaScript</li>
    <li>React</li>
</ul>
```

Solution:

```
let list = document.getElementById("list");
let newItem = document.createElement("li");
newItem.textContent = "CSS";
let secondItem = list.children[1];
list.insertBefore(newItem, secondItem);
```

Output:
HTML
CSS
JavaScript
React

Section 4 - Removing and Cloning Elements

14. Remove an Element

Question:
Create a list containing three items and remove one item using either removeChild() or remove().
HTML:

```
<ul id="list">
    <li>HTML</li>
    <li id="css">CSS</li>
```

```
    <li>JavaScript</li>
</ul>
```

Solution:

```
let cssItem = document.getElementById("css");
cssItem.remove();
```

Output:

HTML
JavaScript

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15. Clone an Element Using cloneNode()

Question:

Create a button and use cloneNode() to create a copy of the button. Add the cloned button to the webpage.

HTML:

```
<div id="container">
    <button id="myButton">Click Me</button>
</div>
```

Solution:

```
let button = document.getElementById("myButton");
let clonedButton = button.cloneNode(true);
let container = document.getElementById("container");
container.append(clonedButton);
```

Output:

[Click Me] [Click Me]